

Mentoring Plan

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NSF AGS-PRF Application

Mentoring Framework

Although the AGS-PRF applicant is the postdoctoral fellow rather than a hired postdoctoral researcher, the fellowship includes a structured mentoring relationship with the sponsoring scientist. The proposed sponsoring scientist is David Keellings, University of Florida, and the proposed host institution is University of Florida, Department of Geography. The mentoring framework will support scientific independence, technical growth, publication planning, professional networking, teaching development, and preparation for a faculty or research-scientist career.

Research Mentoring

The Fellow and sponsoring scientist will meet regularly to review research progress, model diagnostics, physical interpretation, manuscript development, conference abstracts, and reproducible workflow milestones. Mentoring will emphasize rigorous climate-science interpretation of AI results, transparent uncertainty communication, and responsible framing of experimental forecast products.

Professional Development

Professional development will include manuscript planning, proposal-writing feedback, conference presentation preparation, networking with atmospheric-science and geospatial communities, and opportunities to mentor students in climate-data science. The Fellow will also develop teaching material on trustworthy AI for climate extremes and will seek feedback on communicating uncertainty to interdisciplinary audiences.

Assessment

Progress will be assessed through quarterly milestones: baseline and CFM model completion, verification diagnostics, physical-regime analysis, manuscript drafts, public release preparation, and career-development activities. The mentoring relationship will be adjusted as needed to ensure the Fellow develops an independent research identity.